SQL Worksheet – Class 12 – Computer Science

1. Consider the following table **SHOP**. Write SQL commands for the questions (i) to (v).

Item_No	Item_Name	Price	Qty	Total
1021	Hamam	24	2	48
987	Colgate	10	2	10
3623	Surf	52	1	52
277	Shampoo	2	5	10
4855	Face wash	57	1	57

i. To display all the items whose Price is greater than 50.

SELECT * FROM SHOP WHERE Price>50;

ii. To increase the Total of Item_No 987 as 20 (Price*Qty).

UPDATE SHOP SET Total=Price*Qty WHERE Item_No=987;

iii. To list all the Item_Name and Qty in ascending order of Item_No whose Qty is less than 3.

SELECT Item_Name,Qty FROM SHOP WHERE Qty<3 ORDER BY Item_No ASC;

iv. To add an extra column Discount which accepts decimal numbers.

ALTER TABLE SHOP ADD Discount FLOAT;

v. To delete the row(s) whose quantity is 1.

DELETE FROM SHOP WHERE Qty=1;

2. Consider the following table **EMPLOYEE**. Write SQL commands for the questions (i) to (v).

Eid	Ename	Qual	Gender	Salary
901	Kumar	BE	M	12000
201	Louie	MCA	M	30500
110	Varshini	BTech	F	13025
718	Drishya	MCA	F	11600
120	Nissi Ann	MCA	F	19000

i. To display Eid and Ename whose Qual is MCA.

   SELECT Eid,Ename FROM EMPLOYEE WHERE Qual="MCA";

ii. To list the employees in descending order of their Salary.

   SELECT * FROM EMPLOYEE ORDER BY SALARY DESC;

iii. To display the count of the employees according to their Gender.

   SELECT GENDER, COUNT(GENDER) FROM EMPLOYEE GROUP BY GENDER;

iv. To find out the minimum Salary and maximum Salary of the employees.

   SELECT MIN(SALARY),MAX(SALARY) FROM EMPLOYEE;

v. To delete the records whose Salary is above 15000.

   DELETE FROM EMPLOYEE WHERE SALARY>15000;

3. Consider the following table **STUDENT**. Write SQL commands for the questions (i) to (v).

Roll_no	**Stud_name**	**Maths**	**Computer**	**Physics**	**Chemistry**
1201	Hari	48	89	79	99
1202	Karthik	78	100	89	72
1203	Varun	52	90	93	88
1204	Gokul	69	81	75	96
1205	Vignesh	57	100	100	98

i. To increase the Maths mark of all the students by 5.

UPDATE STUDENT SET MATHS=MATHS+5;

ii. To display the Students details whose mark is 100 in Computer.

SELECT * FROM STUDENT WHERE COMPUTER=100;

iii. To display the Name and Roll Number of the Students whose name starts with letter 'V'.

SELECT STUD_NAME, ROLL_NO FROM STUDENT WHERE
STUD_NAME LIKE "V%";

iv. To set Roll Number column as Primary Key.

ALTER TABLE STUDENT MODIFY ROLL_NO INT PRIMARY KEY;

v. To delete the records whose Physics marks are below 85.

DELETE FROM STUDENT WHERE PHYSICS<85;

4. Consider the following table **COLLEGE**. Write SQL commands for the questions (i) to (v).

ID_No	**Name**	**College**	**Branch**	**Roll_No**	**Age**
91	Ramesh	NIT	MECH	8642046	20
72	Jagan	MANIPAL	LAW	8642079	24
64	Kunal	IIT	AEROSPACE	8642011	18
18	Haneefa	IIT	CIVIL	8642054	22
78	Michael	IIT	CSE	8642002	20

i. To display all the students studying at IIT under the age of 20.

   SELECT * FROM COLLEGE WHERE COLLEGE=”IIT” AND AGE<20;

ii. To change the Branch of Name ‘Kunal’ to CSE.

   UPDATE COLLEGE SET BRANCH=”CSE” WHERE NAME=”KUNAL”;

iii. To display the number of students in IIT, NIT & MANIPAL.

   SELECT COLLEGE, COUNT(COLLEGE) FROM COLLEGE

   GROUP BY COLLEGE;

iv. To display the details in ascending order of Roll Number.

   SELECT * FROM COLLEGE ORDER BY ROLL_NO;

v. To delete the records which have Age greater than 20.

   DELETE FROM COLLEGE WHERE AGE>20;

5. Consider the following tables and write SQL commands for the questions (i) to (v).

AUTHORS

author_id	name	nationality	birth_year
1	'George Orwell'	'British'	1903
2	'J.K. Rowling'	'British'	1965
3	'Mark Twain'	'American'	1835
4	'Jane Austen'	'British'	1775
5	'Agatha Christie'	'British'	1890

BOOKS

book_id	title	genre	publish_year	author_id
1	'1984'	'Dystopian'	1949	1
2	'Animal Farm'	'Political Satire'	1945	1
3	'Harry Potter and the Philosophers Stone'	'Fantasy'	1997	2
4	'Harry Potter and the Chamber of Secrets'	'Fantasy'	1998	2
5	'The Adventures of Tom Sawyer'	'Fiction'	1876	3
6	'Pride and Prejudice'	'Romance'	1813	4
7	'Murder on the Orient Express'	'Mystery'	1934	5

i. Display the titles of books along with the names of their authors. Only show books published after 1900.
SELECT TITLE, NAME FROM BOOKS NATURAL JOIN AUTHORS
WHERE PUBLISH_YEAR>1900;

ii. Display the name and nationality of each author along with the titles of the books they have written.
SELECT NAME, NATIONALITY, TITLE FROM BOOKS
NATURAL JOIN AUTHORS;

iii. Count the number of books for each genre.
SELECT GENRE, COUNT(GENRE) FROM BOOKS GROUP BY GENRE;

iv. Show the author's name and the number of books they have written. Find authors who have written more than one book.
SELECT NAME, COUNT(TITLE) FROM BOOKS NATURAL JOIN AUTHORS
GROUP BY NAME HAVING COUNT(TITLE)>1;

v. Retrieve the name of the author and the title of their most recently published book.
SELECT NAME, TITLE, MAX(PUBLISH_YEAR) FROM BOOKS NATURAL JOIN
AUTHORS GROUP BY NAME;

6. Consider the following tables and write SQL commands for the questions (i) to (v).

CUSTOMERS

customer_id	name	email	phone	address
1	'Alice Smith'	'alice@example.com'	'1234567890'	'123 Maple St'
2	'Bob Johnson'	'bob@example.com'	'2345678901'	'456 Oak St'
3	'Carol White'	'carol@example.com'	'3456789012'	'789 Pine St'
4	'David Brown'	'david@example.com'	'4567890123'	'101 Birch St'
5	'Emma Green'	'emma@example.com'	'5678901234'	'202 Cedar St'

ORDERS

order_id	order_date	amount	customer_id
101	'2023-11-01'	250.00	1
102	'2023-11-05'	150.00	1
103	'2023-11-10'	300.00	2
104	'2023-11-15'	400.00	3
105	'2023-11-20'	120.00	2

i. List all customers who placed orders on or after November 10, 2023. Display their name, email, order date, and amount.
ii. Show the total amount spent by each customer. Display the customer's name along with the total amount.
iii. Retrieve all orders, sorted by order amount in descending order. Display the order ID, order date, customer name, and amount.
iv. Find the names of customers who have placed orders with an amount greater than 200. Display the customer’s name and order amount.
v. List customers who have placed more than one order. Show the customer’s name and the number of orders they have placed.

7. Consider the following tables and write SQL commands for the questions (i) to (v).

Table: LIBRARY

Book_ID	Title	Author	Genre	Copies
101	Database Systems	Ramakrishnan	Education	4
102	Operating Systems	Silberschatz	Education	3
103	C Programming	Kernighan	Education	6
104	Harry Potter	Rowling	Fantasy	10
105	Lord of the Rings	Tolkien	Fantasy	2

i. Display the titles of books where Copies are more than 5.
ii. List all Fantasy books sorted by Title.
iii. Count the total number of books by each Genre.
iv. Display Author names who have more than one book in the library.
v. Delete the record of book where Copies = 2.

8. Consider the following tables and write SQL commands for the questions (i) to (v).

Table: BANK

Acc_No	Name	Branch	Balance	City
2001	Amit	Delhi	50000	Delhi
2002	Riya	Mumbai	60000	Mumbai
2003	John	Delhi	25000	Delhi
2004	Meena	Bangalore	80000	Bangalore
2005	Karthik	Mumbai	100000	Mumbai

i. Display Name and Balance of all customers whose Balance is above 50000.
ii. List customers from Mumbai sorted by Balance in descending order.
iii. Find the average Balance of all customers.
iv. Count the number of customers in each City.
v. Update Balance of Acc_No 2003 to 30000.

9. Consider the following tables and write SQL commands for the questions (i) to (v).

Table: HOSPITAL

P_ID	P_Name	Disease	Doctor	Bill
1	Rajesh	Flu	Dr. Kumar	1500
2	Sita	Cancer	Dr. Mehta	20000
3	Rohan	Fracture	Dr. Gupta	12000
4	Priya	Flu	Dr. Kumar	1800
5	Anil	Diabetes	Dr. Singh	5000

i. Display the names of all patients treated by Dr. Kumar.
ii. List all patients whose Bill is greater than 10000.
iii. Display distinct Diseases treated in the hospital.
iv. Show the maximum and minimum Bill amount.
v. Delete all records where Bill < 2000.

10. Consider the following tables and write SQL commands for the questions (i) to (v).

Table: AIRLINES

Flight_No	Airline	Source	Destination	Fare
301	IndiGo	Delhi	Mumbai	4500
302	Air India	Chennai	Delhi	6000
303	Vistara	Bangalore	Delhi	7000
304	SpiceJet	Delhi	Kolkata	5500
305	IndiGo	Delhi	Chennai	5000

i. Display details of flights with Fare above 5000.
ii. List all flights from Delhi sorted by Fare.
iii. Count the number of flights for each Airline.
iv. Display the Airline with the cheapest Fare.
v. Update Fare of Flight_No 301 to 4800.

11. Consider the following tables and write SQL commands for the questions (i) to (v).

Table: CINEMA

Movie_ID	Movie_Name	Director	Genre	Collection
11	Inception	Nolan	Sci-Fi	200
12	Interstellar	Nolan	Sci-Fi	300
13	Avengers	Russo	Action	400
14	Titanic	Cameron	Romance	350
15	Avatar	Cameron	Sci-Fi	500

i. List all movies directed by Nolan.
ii. Display the highest Collection movie.
iii. Count movies according to Genre.
iv. Display the names of movies where Collection > 300.
v. Delete the record of Titanic.

12. Consider the following tables and write SQL commands for the questions (i) to (v).

Table: ONLINE_SHOPPING

Order_ID	Customer	Product	Qty	Amount
901	Amit	Mobile	1	15000
902	Riya	Laptop	1	55000
903	Karan	Headphones	2	4000
904	Meena	Tablet	1	20000
905	Karthik	Laptop	1	60000

i. Display all orders where Amount > 20000.
ii. Show the total Amount spent by each Customer.
iii. List customers who purchased Laptop.
iv. Count the number of orders per Product.
v. Update Qty = 3 for Order_ID 903.

13. Consider the following tables STORE and ITEM and answer (a) and (b) parts of this question.

Table: STORE

SNo	SName	Area
S01	ABC Conputronics	GK II
S02	All Infotech Media	CP
S03	Tech Shoppe	Nehru Place
S04	Geeks Tecno Soft	Nehru Place
S05	Hitech Tech Store	CP

Table: ITEM

INo	IName	Price	Sno
T01	Mother Board	12000	S01
T02	Hard Disk	5000	S01
T03	Keyboard	500	S02
T04	Mouse	300	S01
T05	Mother Board	13000	S02
T06	Key Board	400	S03
T07	LCD	6000	S04
T08	LCD	5500	S05
T09	Mouse	350	S05
T10	Hard Disk	4500	S03

(a) Write the SQL queries ((i) to (iv))

(i) To display IName and Price of all the Items in ascending order of their price.

(ii) To display SNo and SName of all Stores located in CP.

(iii) To display minimum and maximum prices of each IName from the table ITEM.

(iv) To display IName, Price of all items and their respective SName where they are available.

(b) Write the output of the following SQL commands ((i) to (iv)):

(i) SELECT DISTINCT INAME FROM ITEM WHERE PRICE>=5000;

(ii) SELECT AREA, COUNT (*) FROM STORE GROUP BY AREA;

(iii) SELECT COUNT (DISTINCT AREA) FROM STORE;

(iv) SELECT INAME, PRICE*0.05 DISCOUNT FROM ITEM WHERE SNO IN (‘S02’, ‘S03’);

14. Consider the following table ‘SOFTDRINK’. Write SQL commands for the following questions:

Table: SOFTDRINK

Drinkcode	Dname	Price	Calories
101	Lime and Lemon	20.00	120
102	Apple Drink	18.00	120
103	Nature Nectar	15.00	115
104	Green Mango	15.00	140
105	Aam Panna	20.00	135
106	Mango Juice Bar	12.00	150

i. To display the names and drink codes of 101,102,105,106.

ii. To display drink codes, names and calories of all drinks in descending order of calories.

iii. To display names and price of Green Mango and Mango Juice Bar.

iv. To increase the calories of all drinks by 10%.

v. To display the maximum, minimum calorie drink and sum of price of all drinks in the given table.

15. Consider the following table ‘GYM’. Write SQL commands for the following questions:

Table: GYM

ICODE	INAME	PRICE	BRAND
G101	Power fit exerciser	20000.00	Power Gymea
G102	Aqua fit hand grip	1800.00	Reliable
G103	Cycle bike	14000.00	Eco bike
G104	Protoner extreme gym	3000.00	Cosco
G105	Massage belt	5000.75	Massage expert
G106	Cross trainer	13000.50	GTC fitness

i. To display the names of all the items whose name starts with ‘A’.
ii. To display the ICODE and INAME of all the items whose brand name is Reliable or Cosco.
iii. To change the brand to ‘Fit trend India’ of the item whose ICODE is G101.
iv. Add a new row in GYM table with following details:
‘G107’, ’Exerciser’, 21000, ’GTC Fitness’
v. Create query to create this table GYM.

16. Consider the following table ‘FITNESS’. Write SQL commands for the following questions:

Table: FITNESS

Pcode	ICODE	Pname	Price	Manufacturer
P1	G101	Treadmill	21000	Cosco
P2	G102	Bike	20000	Aone
P3	G103	Cross trainer	14000	Reliable
P4	G104	Multi gym	30000	Cosco
P5	G105	Massage chair	5500	Regrosene
P6	G106	Belly vibrator belt	9000	Ambawya

i. To display the names of all the products with price more than 20000.
ii. To display the names of all products by the manufacturer Cosco.
iii. To change the price data of products whose price is greater than 20000 with discount reduction of 25%.
iv. To display the name and manufacturer of the products whose names starts with ‘B’ and ends with ‘E’.
v. To display the average price of all the products.